

HOME AUTOMATION ANDROID APPLICATION USING ARDUINO

**A System Analysis and Design Project
Presented to the Faculty of
College of Information Technology Education
GOLDEN WEST COLLEGES
Alaminos City, Pangasinan**

**In Partial Fulfillment of the Requirements
in Degree Program
Bachelor of Science in Information Technology**

**Research By:
Calonge, Jehosaphat D.
Pioquinto, Jeremy L.
Cardenas, Angel R.
Tagara, Phinkie R.
Reyes, Matthew B.
Dayrit, Melanie V.
Bravo, Jericho P.**

**Adviser:
John Vianney V. Manuel**

November 2023

GOLDEN WEST COLLEGES

San Jose Drive, Alaminos City, Pangasinan *Tel No. (075) 552-7382

APPROVAL SHEET

This System Analysis and Design Project entitled "HOME AUTOMATION ANDROID APPLICATION USING ARDUINO" prepared by Jehosaphat D. Calonge, Jeremy L. Ploquinto, Angel R. Cardenas, Phinkie R. Tagara, Matthew B. Reyes, Melanie V. Dayrit, Jericho P. Bravo in partial fulfillment of the requirements in the Bachelor of Science in Information Technology has been examined and is recommended for approval and acceptance this November 2023.

JOHN VIANNEY V. MANUEL
System Analysis and Design, Adviser

THE ORAL EXAMINATION COMMITTEE

Oral Examination Grade _____

KAIROS JIL I. MOLINA
Chairman

MARK LESLIE D. MELENDEZ
Member

LUIELYN B. RAGUINDIN
Member

APPROVED and ACCEPTED in partial fulfillment of their requirements in Bachelor of Science in Information Technology.

KAIROS JIL I. MOLINA
Dean, CITE Department

ACKNOWLEDGEMENT

We, the authors of this research paper, would like to express our sincere gratitude to the individuals and organizations that have contributed to the successful completion of this project.

To the researcher's parents/guardians, Mr. Ronaldo M. Dayrit, Mr. Adan Cesar E. Reyes, Mr. Fred B. Tagara, Mr. Romano G. Cardenas, Mr. Florencio R. Calonge, Mr. Larry C. Bravo, Mr. Allen P. Pioquinto for their unwavering financial support and encouragement. Their belief in our abilities has motivated us to work hard and persevere through the challenges of this project. We are grateful for their guidance and encouragement, which have been instrumental in our academic success.

To our adviser, Mr. John Vianney V. Manuel for their invaluable guidance and expertise. Their knowledge and experience have been a great asset to our research, and we are fortunate to have had their mentorship throughout this project.

To our friends and classmates who have provided us with encouragement, feedback, and assistance throughout the research process. Their contributions have been invaluable, and we could not have completed this project without their help.

In conclusion, we would like to thank all those who have supported us in this endeavor, and we hope that our research will

DEDICATION

This research paper is sincerely dedicated to the developer's parents and guardians, Mr. Larry C. Bravo, Mr. Florencio R. Calonge, Mr. Adan Cesar E. Reyes, Mr. Romano G. Cardenas, Mr. Ronaldo M. Dayrit, Mr. Allen P. Pioquinto, and Mr. Fred B. Tagara, who encouraged and inspired the developers in conducting the study. They never left their side throughout the process, which gave them strength and hope when they thought of giving up. They gave the developers a great sense of enthusiasm and perseverance in continuing this study. Without their love and assistance, this research would not have been possible.

The developers want to dedicate this research work to Mr. John Vianney V. Manuel for being supportive as the adviser of the System Analysis and Design Project. The developers appreciate your words of advice and encouragement in finishing the study and in continuously giving moral and emotional support.

The developers also want to dedicate this research work to Mr. Mark Leslie D. Melendez for his guidance and teaching to make the study better. The developers appreciate your words of advice and encouragement in finishing the study and in continuously giving moral and emotional support.

The developers also want to dedicate this research work to Ms. Luielyn B. Raguindin for her guidance and teaching to make the study better. The developers appreciate your words of advice and encouragement in finishing the study and in continuously giving moral and emotional support.

And lastly, the developers dedicated this research paper to the Almighty God, who gave them strength, wisdom, guidance, power of thinking, security, competence, and for giving good health while doing this study.

- The Developers

TABLE OF CONTENTS

Title Page	i
Approval Letter	ii
Acknowledgement	iii
Dedication	v
Table of Contents	vii
List of Figures	ix
Chapter 1: Introduction	1
Background of the Study	1
Conceptual Framework	3
Statement of Objectives	6
Significance of the Study	6
Scope and Limitations	7
Definition of Terms	9
Chapter 2: Review of Related Literature	11
Local Literatures	11
Foreign Literatures	15
Chapter 3: Design and Methodology	20
Research Design	20
Software Developmental Model	21

Sources of Data 24

Implementation and Data Collection25

Tools for Data Analysis 26

Requirements Analysis29

Requirements Documentation 33

Implementation Plan44

Chapter 4: Summary, Conclusions, Recommendations 46

Summary46

Conclusion48

Recommendation48

References 50

Appendices 55

Appendix A: Survey Questionnaire 55

Appendix B: Sample Survey56

Appendix C: Gantt Chart59

Appendix D: Source Code60

Appendix E: Parts of the Developed System 61

Curriculum Vitae68

LIST OF FIGURES

Figure 1. Research Paradigm	5
Figure 2. Rapid Application Development Life Cycle	22
Figure 3.1 Opening the Garage Door	29
Figure 3.2 Closing the Garage Door	29
Figure 3.3 Switching the Light On	30
Figure 3.4 Switching the Light Off	30
Figure 3.5 Switching the Ceiling Fan On	31
Figure 3.6 Switching the Ceiling Fan Off	31
Figure 3.7 Switching the Household Appliances Off	32

CHAPTER 1

Introduction

Background of the Study

In this modern world, automation technology is one of the most essential and appealing fields that plays a significant part in daily life. Automation is a strategy, method, or system for operating or regulating a process with electronic equipment in order to reduce the need for human intervention. The principles of constructing an automation system for a business or home provide numerous advantages. Industrialists and researchers are attempting to develop efficient and cost-effective automated systems to monitor and manage various machinery, such as lights, fans, and air conditioners, as needed. Automation not only enables more efficient and waste-free use of power and water, automation is another key application of wireless technology, such as bluetooth. It involves the monitoring and control of the consumption of energy and the environment in buildings, educational institutions, workplaces, and museums. Low-cost communication technology such as bluetooth assists in making it more operative and efficient (Gaikwad & Kalshetty, 2017).

Home Automation Android Apps using Arduino can be controlled remotely, which is convenient and efficient for disabled persons, elders that have good eyesight, and other individuals that can control the

device. It only needs a smartphone device and bluetooth so that it can operate the application. It also saves time and money. It is easy to assemble home appliances, and the most significant is that there is no risk of electric shock (Vijayvrgiya et. al., 2016).

The client will use a smartphone device for operating the application, and it needs bluetooth as the connection; however, it has only approximately 30 feet of bluetooth as a connection, which is limited depending on the type of bluetooth device. It cannot detect the system if the client is distant from the required range. The smartphone should have a good amount of battery so that the client will have a good connection. Increasing safety, convenience, control, comfort, and energy savings makes our automation better than the existing automations (Malav et. al., 2019).

The purpose of the Home Automation Android App using Arduino is to make it possible to automate tasks related to well-being. It helps to ease daily life by increasing user comfort. It will monitor and/or control the consumption of energy, which can lower the user's monthly bills and potentially even reduce their insurance premiums. It also helps to manage the home devices with the use of a smartphone as the remote control in one place. The need for automation has been increasing a lot, and automation holds more potential than it is currently being used for. As technology advances, new ways to use automation and its applications will also vastly increase. Automation

can be used to make homes friendlier for the elderly and children (Singh et. al., 2021).

Home automation has established itself as the go-to solution for individuals seeking a dependable and user-friendly platform to control and automate their smart home devices, which transform a traditional house into smarter, more productive, and more secure living spaces. Users can remotely control various aspects of their home environment, including lighting and entertainment.

Home automation aims to simplify daily life, promote sustainability, and combine hardware and software with accepted styles of features into a single, organized ecosystem.

Conceptual Framework

The Input-Process-Output model (IPO) served as the conceptual framework that the developers utilized. The project's general structure and guiding principles are provided by the IPO model.

Figure 1 illustrates the objectives of the proponents, which include saving proper energy management of electricity distribution, developing a wireless home automation system controlled by a smartphone, specifically an android device, creating and executing a reliable, cost-effective home automation system, and providing a user-

friendly and safe system to manage home appliances, with a focus on assisting the elderly and disabled persons.

These inputs were transformed into their expected output using the Agile Development Scrum, which comprises six phases: plan, design, develop, test, deploy, and review. The result was the Home Automation Android App using Arduino.

The Figure 1 on the next page shows the research paradigm of the study.

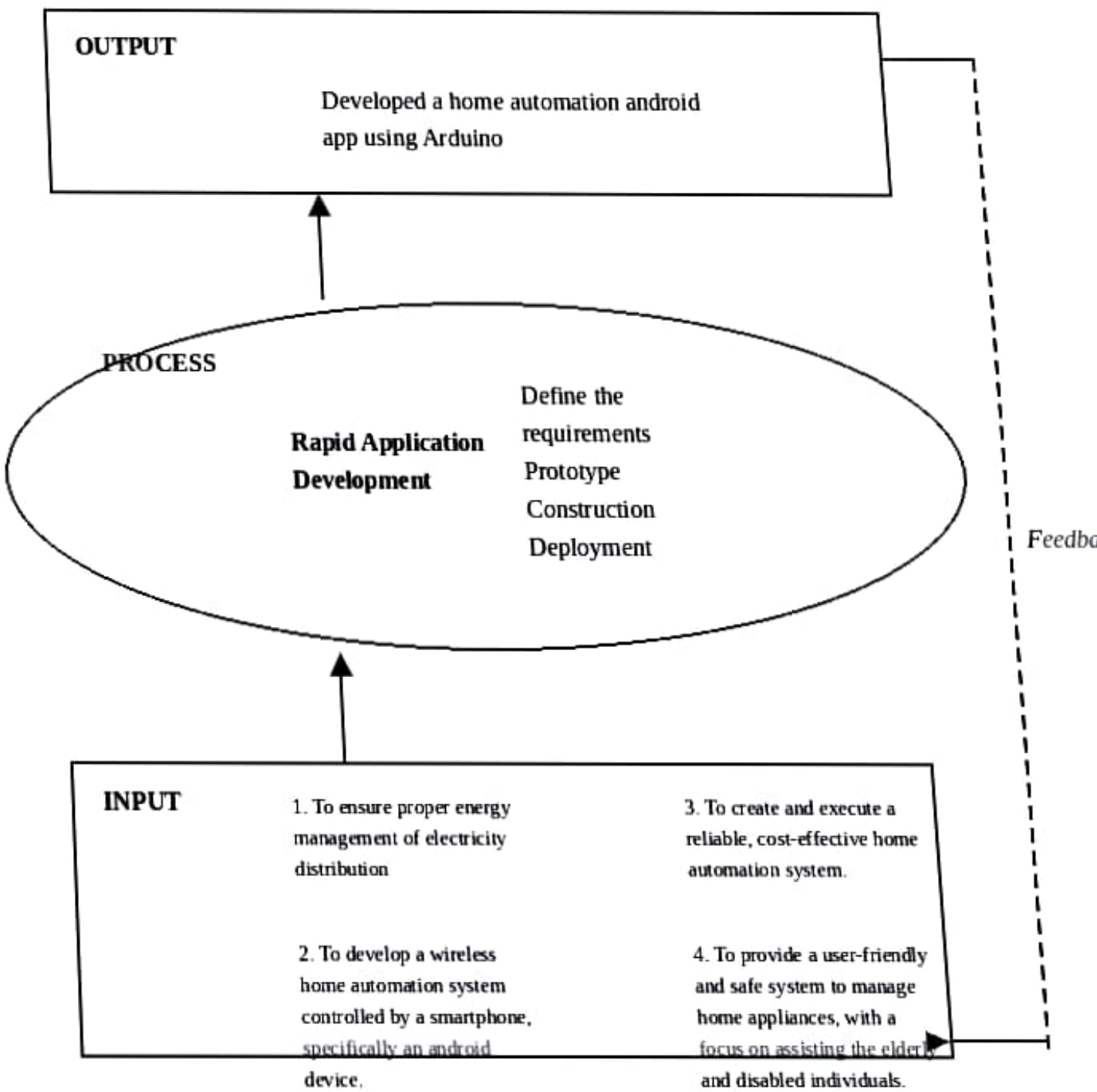


Figure 1. Research Paradigm

Statement of Objectives

The purpose of this project is to develop a home automation Android app using Arduino that can control some appliances like electric fans, lights, etc. to prevent the inconvenience of disabled persons, elders that have good eyesight, and other individuals that can control the device and easily access and control the home appliances by staying at a particular place and accessing them remotely without the help of other people.

Specifically, the developers aim to achieve are the following:

1. To ensure proper energy management of electricity distribution,
2. To develop a wireless home automation system controlled by a smartphone, specifically an android device,
3. To create and execute a reliable, cost-effective home automation system, and
4. To provide a user-friendly and safe system to manage home appliances, with a focus on assisting the elderly and disabled individuals.

Significance of the Study

The focus of the developers is to develop an automation that can help to enhance convenience and comfort, reduce energy consumption and lower utility bills, make the application easily accessible, and lead

to cost savings through energy efficiency and reduced maintenance charges. The beneficiaries of the study are the following:

Homeowners. Home automation apps offer convenience and control over different components of a home, like lights, fans, and so on. Homeowners can improve their living environment while saving electricity.

Elderly or Disabled Individuals. Home automation can make daily duties easier for people who have restricted mobility or disabilities. Remote control via an Android app can help you turn on and off lights, fans, and other devices.

Developers. For the developers, this study will give them more knowledge and skills.

Future Researchers. This study can be helpful for future researchers, and they can use it for their related studies.

Scope and Limitations

The scope of the study is to control a wide range of devices in your home, such as lights, fans, and garage doors. The users can use the app to turn devices on and off, which is a great way to control the user's home, which can be convenient for the user. They can use the application to control the following appliances: turning on the lights

and the garage door when arriving at home and turning off lights and fans when leaving home.

The limitation of the study is that it requires some technical knowledge when using the devices. It is also necessary to be able to wire and connect the Arduino to the devices and to be able to write code to control the Arduino.

The second limitation is that the range of the Arduino is limited. It means that the user needs to be close to the Arduino to be able to function with it or control it. The connection of the bluetooth module to the smartphone is limited to one house.

Another limitation is that the connection to the bluetooth module is limited to one user at a time. Bluetooth modules enforce a single connection unit, which involves rejecting additional connection requests while the connection is already established.

Finally, it is also important that there is always a possibility that something could go wrong, such as a power outage or a software glitch, which causes the devices to malfunction or become unresponsive and cannot always be reliable.

Definition of Terms

Android App. An Android app is a software application designed to run on Android-powered smartphones and tablets. It serves as the user interface for controlling and monitoring home automation devices.

Arduino. Arduino is an open-source electronics platform that consists of both hardware and software components. It is commonly used for building DIY electronic projects, including home automation systems. Arduino boards are programmed to interact with sensors, actuators, and other hardware components.

Bluetooth Module. These modules enable communication between the Android app and the Arduino board. A bluetooth module (e.g., HC-05/HC-06) can be used for short-range communication.

Home Automation. Home automation refers to the use of technology to control and automate various household tasks and functions. In the context of your project, it involves controlling lights, appliances, security systems, and other devices remotely or automatically through your Android app.

Micro-controller. The micro-controller on the Arduino board is responsible for processing instructions and controlling connected devices.

Relay. A relay is an electrically operated switch that can be turned on or off, letting the current go through or not, and can be controlled with low voltages, like the 5V provided by the Arduino pins.

User Interface (UI). The UI of your Android app is what users interact with. It should provide a user-friendly interface for controlling and monitoring home automation devices. This includes buttons, sliders, switches, and status indicators.

CHAPTER 2

Review of Related Literatures

Local Literatures

According to Morallo (2021), bluetooth is now mainstream and synonymous with the 21st century. Wireless technology enables the exchange of data over short distances using shortwave radio transmission, providing convenience, intelligence, and control. In her article, she proposed and prototyped a home lighting control system that uses an Android smartphone as a platform and an Arduino bluetooth interface. Check out Arduino, Android smartphones, and bluetooth technology. This design uses the Bluetooth SPP (Serial Port Protocol) HC-05 module, which is designed to transparently establish wireless serial connections. The Android application is downloaded onto an Android smartphone and remotely monitors and controls the operation of the lighting system. This application allows you to easily control the operating pins of your Arduino. This design shows that the Arduino bluetooth interface can control the lighting in your home using an Android smartphone as a platform. Future lighting home automation will include more energy-efficient and scalable timers. GSM modules will be integrated into future designs of secure, universally accessible, and remotely controlled lighting systems. It will also be

expanded to larger environments, such as universities, offices, and factories.

According to Techatronic (2020), there are many possibilities for home automation, such as wi-fi-based home automation, Web-based home automation, and IoT control house automation, but this time we're posting the simplest in every category so even novices may create this amazing project. Automation refers to the ability to operate every appliance linked to the system using a bluetooth-enabled smartphone app. MIT App Inventor can also be used to create this application.

According to Caldo et al. (2015), the goal of their project is to create a prototype that can be used in conjunction with an Android phone and a local area network (LAN) or the internet to operate as a centralized switch for basic household appliances. With a master switch, the prototype can manage up to seven devices simultaneously. The Arduino microcontroller in the prototype's main body is wired to a relay driver circuit. The Basic4Android IDE is also used to develop an Android application. Its supporters include features like password security and device renaming to create an interface that is easy to use. To ascertain whether the prototype satisfies the scope and limitations, the proponents employed the developmental method and carried out several tests. The prototype was tested by the creators to support loads ranging from two to seven output devices and demonstrated that the prototype is capable of automating the switching of basic household

appliances, paving the way for the development of a complete home automation system.

According to Ngerem et al. (2021), their study aims to examine and create a basic home automation system. This is an idea that allows devices to share data and be controlled remotely from a central hub. It is suggested to use bluetooth technology in the design and construction of a home automation system. The adoption of bluetooth is due to its low cost and ease of accessibility as a network. A circuit design for the devices' operation and software design for the Android smartphone are part of the implementation.

According to Cruz et al. (2020), building a "smart home" or "smart house" is what home automation entails. Appliances, lighting, and entertainment systems can all be managed by a home automation system. Using an Android application, the suggested idea automates a home with MCU technology. Most electrical equipment in homes is manually turned on and off using switches. The project's proponents wanted to automate homes by allowing users to use an Android application to control their appliances. The user can reduce the amount of time they spend manually turning on and off their appliances by using this device.

According to Cruz et al. (2016), modern home designs are starting to follow the trend of integrating home automation systems. These systems might have the ability to remotely switch appliances or devices.

Various technologies are employed to accomplish this purpose for homes dubbed "intelligent homes." A basic model for a remote switching application system has been developed by the proponents using the relevant studies that they have gathered and further examined. The advocates decided to create a model with a phone system. To transmit instruction signals, a telephone line is selected as the channel. There will be a local section and a remote section that make up the system. The common dialing method for telephone systems, DTMF signaling, will be utilized as a method of sending signals to the local section. A DTMF decoder will be used to interpret the signals produced by the control section. An Arduino microcontroller is the system's central nervous system. This will handle the system's primary functions' processing.

According to Dioses Jr. (2020), the days of smart houses are long gone in the modern world of today. Home automation has achieved several milestones and recognition in the last few decades, mostly thanks to science fiction films. Home automation has given life a whole new meaning when compared to earlier times. Electronic gadgets used in homes, such as televisions, air conditioners, lights, electric fans, etc., are managed by controllers.

The arduino bluetooth interface can control the lighting in your home using an Android smartphone as a platform. The Android application is downloaded onto an Android phone and remotely

Various technologies are employed to accomplish this purpose for homes dubbed "intelligent homes." A basic model for a remote switching application system has been developed by the proponents using the relevant studies that they have gathered and further examined. The advocates decided to create a model with a phone system. To transmit instruction signals, a telephone line is selected as the channel. There will be a local section and a remote section that make up the system. The common dialing method for telephone systems, DTMF signaling, will be utilized as a method of sending signals to the local section. A DTMF decoder will be used to interpret the signals produced by the control section. An Arduino microcontroller is the system's central nervous system. This will handle the system's primary functions' processing.

According to Dioses Jr. (2020), the days of smart houses are long gone in the modern world of today. Home automation has achieved several milestones and recognition in the last few decades, mostly thanks to science fiction films. Home automation has given life a whole new meaning when compared to earlier times. Electronic gadgets used in homes, such as televisions, air conditioners, lights, electric fans, etc., are managed by controllers.

The arduino bluetooth interface can control the lighting in your home using an Android smartphone as a platform. The Android application is downloaded onto an Android phone and remotely

Android-based smartphone's user-friendly interface makes the system easy to use. Testimonials indicate that the system makes it easier for the targeted users—elderly and disabled people—to operate lights, heating, air conditioning, and security systems in their homes.

According to Selvaraj et al. (2016), in their article, they use bluetooth & android OS to create a smart home. Their difficulty in controlling the appliances can be made easier by the proposed Home Automation System that has a remote module for controlling the appliances. A remote-controlled home automation system meets the needs of people. The remote-control operation is done by a smartphone/tablet that supports Android OS. The smartphone/tablet acts as a transmitter and sends commands to the receiving bluetooth module.

According to Rosendahl et al. (2007), their paper examines mobile home automation, a domain that arises from the amalgamation of home automation technologies and mobile application platforms. They first introduce a two-dimensional conceptual model of mobility in construction to highlight the necessity for such applications. Next, they propose an architecture and go over various ways a user could access the controlled devices and a mobile home automation service. They also developed a service for mobile home automation that they called REMOTILE. This prototype facilitates our discussion of common

elements, like modules that combine different home automation devices.

According to Hafiz (2020), the world is rapidly approaching automation. Since people don't have as much time as they used to, automation is a quick and easy approach to control any gadget to our specifications. The goal is to use Arduino to construct and design a home automation system. Home automation systems that use an Arduino Uno with a bluetooth module operate household appliances including fans, lights, air conditioners, and automatic door locks. The primary focus is on using an Android phone to monitor and control a smart home and to provide a security-based smart home when the owners are away from home. The goal of this article is to control household appliances in smart homes that are easy to use, affordably designed, and require little installation.

According to Malik et al. (2022), the world's living standards have seen a gradual shift due to the swift advancement of technology. An automated home control system has grown in value as a result of the rising number of closed-source gadgets found in the homes. However, the complexity, high cost, closed-source nature, and incompatibility of numerous standards in current systems lead to limited home automation adoption among wealthy individuals or enthusiasts. The primary goal of this effort is to create a home automation system that can be remotely managed by any Android OS smartphone utilizing an

Arduino board equipped with bluetooth. Homes are becoming increasingly intelligent as technology advances. Conventional switches in modern homes are rapidly being replaced with centralized control systems with remote-controlled switches.

According to Piyare et al. (2011), a low-cost, flexible, and secure cellphone-based home automation system is presented in the paper. The system is based on a stand-alone Arduino BT board, and home appliances connected to its input/output ports via relays. The communication between the Arduino BT board and the cell phone is wireless. The system is designed to be low-cost and scalable, allowing a variety of devices to be controlled with minimal changes to its core. Technology is an ever-evolving process. The ability to design a product using current technology that will improve the lives of others is a huge contribution to the community. The appliances at home are password-protected so that only authorized individuals can access them.

The project describes how to implement a low-cost home automation system while adhering to assistive technology guidelines. The system is intended for use by the elderly and those with disabilities. The Android-based smartphone's user-friendly interface makes the system easy to use. The remote-control operation is done by a smartphone/tablet that supports Android OS.

The goal of this article is to control household appliances in smart homes that are easy to use, affordably designed, and require little installation. The primary focus is on using an Android phone to monitor and control a smart home and to provide a security-based smart home when the owners are away from home. The goal is to use Arduino to construct and design a home automation system.

A low-cost, flexible, and secure cellphone-based home automation system is presented in this paper. The system is based on a stand-alone Arduino BT board, and home appliances are connected to its input/output ports via relays. The communication between the Arduino board and the cell phone is wireless.

CHAPTER 3

Methodology

Research Design

Research Design is a method for utilizing empirical data in response to the subject of study. Furthermore, it refers to the comprehensive method used to conduct research that establishes a clear and logical plan to address the research questions through the gathering, interpreting, analyzing, and debating of data.

According to Creswell J.W and Creswell, J.D (2017), a research design refers to the overall plan or structure that guides a research study. It outlines the framework and methodology that researchers will use to collect, analyze, and interpret data in order to answer their research questions or achieve their research objectives. A well-thought-out research design is crucial for ensuring the validity, reliability, and relevance of research finding

According to Johnson and Christensen (2019), that a research design emphasizes that the study design will be determined by a variety of elements, including the research topic, the type of data available, and the resources available to the researcher. It is essential to choose a research design that is appropriate for the study topic and allows the researcher to collect and assess data in a fair and systematic manner.

According to Leedy and Ormrod (2018), a well-designed research study will be systematic and objective, and it will minimize the potential for bias. It will also be feasible and ethical, and it will be appropriate for the specific research question or hypothesis being studied.

Software Developmental Model

Rapid Application Development, or RAD, means an adaptive software development model based on prototyping and quick feedback with less emphasis on specific planning. In general, the RAD approach prioritizes development and building a prototype, rather than planning. With rapid application development, developers can quickly make multiple iterations and updates to the software without starting from scratch. This helps ensure that the final outcome is more quality-focused and aligns with the end users' requirements.

According to Nalendra (2021), RAD emphasizes designed development cycle much faster with higher quality results than those achieved with other development methodologies. In RAD less emphasis on the task of planning and an emphasis on further development.

Rapid Application Development (RAD)

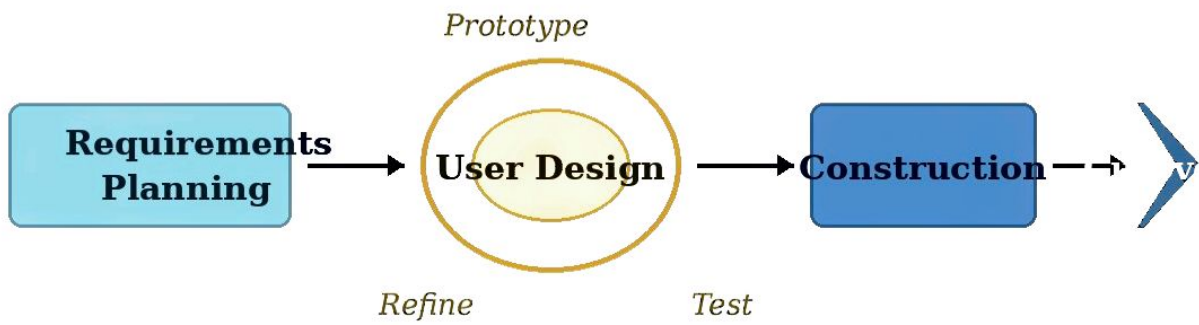


Figure 2. Rapid Application Development Life Cycle

The Stages of Rapid Application Development

Define the requirements. At the very beginning, rapid application development sets itself apart from traditional software development models. It doesn't require you to sit with end users and get a detailed list of specifications; instead, it asks for a broad requirement. The broad nature of the requirements helps you take the time to segment specific requirements at different points of the development cycle.

The developers will collect the required data from the survey. In this stage, the developers will get some information about the necessity of having a home automation in every home.

Prototype. This is where the actual development takes place. Instead of following a rigid set of requirements, developers create prototypes with different features and functions as fast as they can.

These prototypes are then shown to the clients who decide what they like and what they don't.

More often than not, these prototypes are quickly made to work to showcase just the key features. This is normal, and the final product is only created during the finalization stage where the client and developer are in alignment with the final product.

In this stage, the developers will work hand-in-hand until the developed application and automation is ready.

Construction. The construction stage is a crucial stage of development. Engineers and developers work tirelessly to flesh out a working system from a working model. Feedback and reviews are crucial at this stage and most bugs, issues, and alterations are addressed during this stage. This stage can be particularly long, especially in cases where clients change directions or feedback is intensive.

In this stage, the developers will show to the clients about the home automation and receive feedback from the client about the effectiveness and efficiency of the home automation.

Deployment. The final stage of RAD involves deploying the built system into a live production environment. The deployment phase involves intensive scale testing, technical documentation, issue tracking, final customizations, and system simulation. Teams also

Instrumentation and Data Collection

This section defines the processes that the developers used to analyze whether they have done it correctly.

This project aims to provide an automatic home automation Android app using Arduino. It enables the automation of tasks related to safety, health and comfort through intelligent system installed in home and buildings. In other words, integrating technology into special planning. One of the most important advantages of Home Automation systems is energy efficiency.

Survey. A survey is a method of collecting data or information from a group of individuals. It involves asking a series of questions or gathering feedback from respondents. The purpose of a survey is to obtain insights, opinions, or experiences from the target audience in order to analyze and draw conclusions from the collected data. Surveys can be conducted through various methods, such as online questionnaires, face-to-face interviews, telephone interviews, or paper-based forms. The data collected from surveys can be used to make informed decisions, conduct research, evaluate customer satisfaction, measure public opinion, or gather feedback.

The developers conducted a survey to learn more information and learning the effectiveness and efficiency of the home automation.

Library Research. The act of collecting information and conducting investigations by utilizing the various resources found within a library.

The developers were able to do library research and obtain further information that is related to the study.

Online Research. Online research provides fast and convenient access to a wide array of information, such as articles, books, websites, and various forms of multimedia.

The developers mostly find more information online and collect more data from different articles that are related to the study.

Tools for Data Analysis

The developers used the following tools to collect data for the creation of the proposed study. These collected data are analyzed to understand the information that relates to the study.

Data Set – a data set is an ordered collection of data. As we know, a collection of information obtained through observations, measurements, study, or analysis is referred to as data. It could include information such as facts, numbers, figures, names, or even basic descriptions of objects.

The developers utilize datasets to enhance the system's capabilities.

Database Schema - a database schema refers to the logical and visual configuration of the entire relational database. The database objects are often grouped and displayed as tables, functions, and relations. A schema describes the organization and storage of data in a database and defines the relationship between various tables.

It is an advantage for developers to organize and understand the structured data or database.

Entity Relationship Diagram - it is a data modeling technique that graphically illustrates an information system's entities and the relationship between the entities. An ERD is a conceptual and representational model of data to represent the entity framework infrastructure.

It will help the developers to simply have a perfect view of all database elements like setting the values of all the attributes, types of relationships, and other criteria.

Flow Chart - a flow chart is a graphical representation of a process or system that uses different shapes, symbols, and arrows to depict the flow of steps, actions, and decisions within that process. Flow charts are commonly used in various field, including computer programming, engineering, business processes, and data analysis to visually illustrate how something works or is carried out.

A flow chart visually represents the logical flow of a program or process, adding to understanding, debugging, and improving the efficiency of the code or workflow, which helps the developers.

Use Case Diagram - A use case diagram is a depiction of how interactions and connections within a system interact. These illustrations give a comprehensive overview of a system. It might display business processes, computer software systems, or client experiences. A use case diagram displays a sample scenario in which users interact with a system using a variety of specialized connectors and symbols.

The use case diagrams created by developers will outline how the system will function in various scenarios.

Weighted Average Mean - The weighted average mean is a type of mean calculated by multiplying the weight associated with a particular event or outcome with its associated quantitative outcome and then summing all the products together. In other words, when some values weigh more than others, the weighted average mean is calculated.

The developers will use weighted average mean to represent the average of a given data.

Gantt Chart - A Gantt chart is defined as a graphical representation of activity against time; it helps project professionals monitor progress.

Requirement Analysis

To begin with, the client will locate the garage button and need to press it to operate the garage door manually. Once the button is pressed, it will automatically open the garage door until it reaches the desired height or until it is fully open.

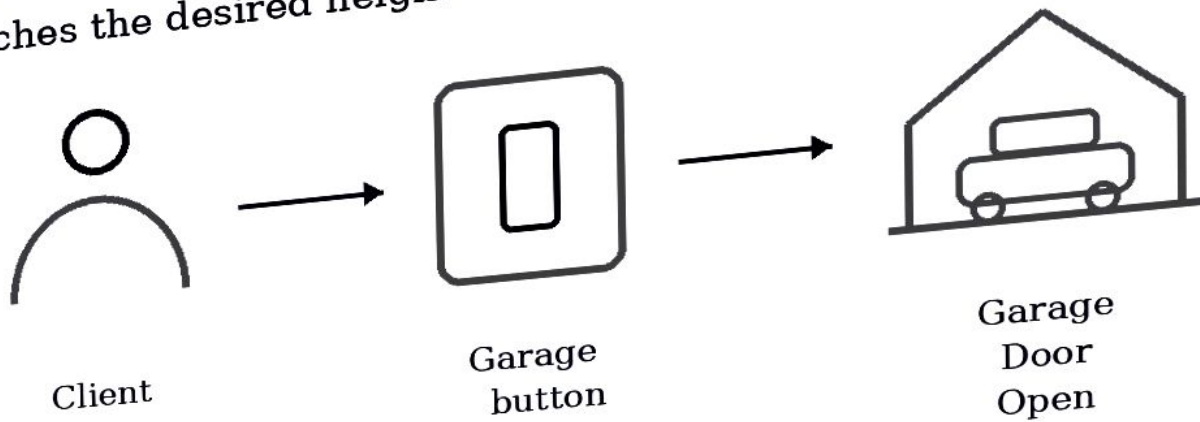


Figure 3.1 Opening the Garage Door

If the client wanted to close the garage door, the client will locate the garage button and need to press and hold the button to operate the garage door manually. Once the button is pressed, it will automatically close the garage door until it is fully closed.

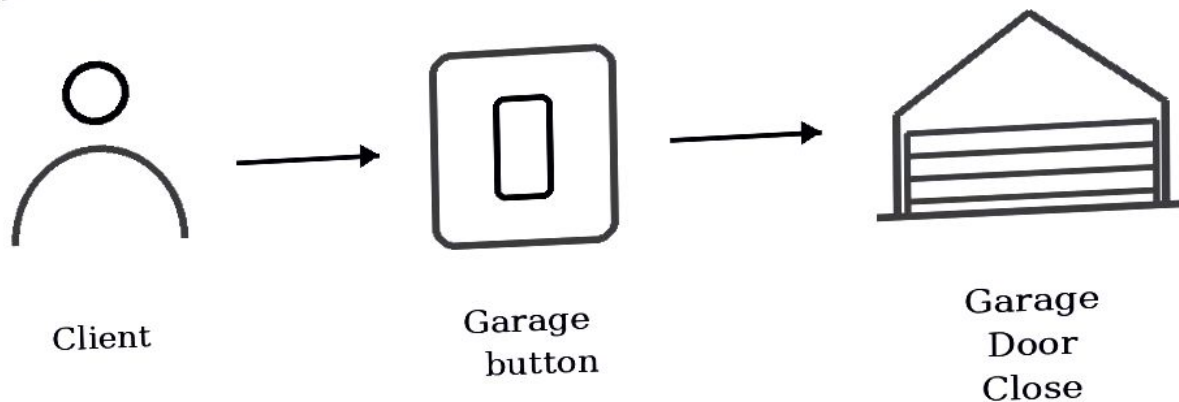


Figure 3.2 Closing the Garage Door

If the client wants to turn on the light, the client will locate the light button and press the button to turn on the light manually. Once the button is pressed, the light will turn on.

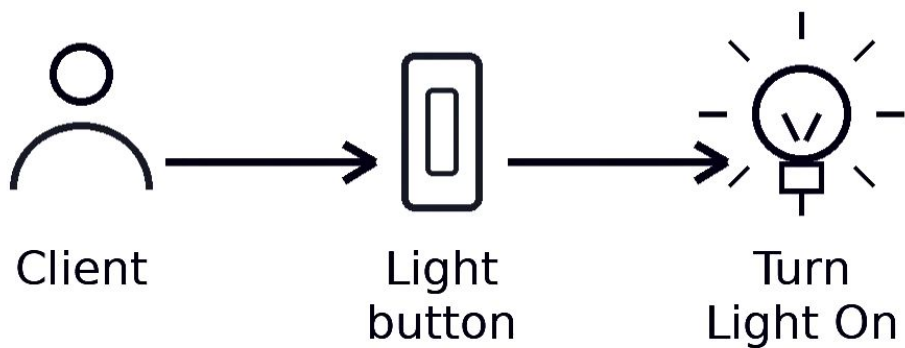


Figure 3.3 Switching the Light On

If the client wants to turn off the light, the client will locate the light button and press the button to turn off the light manually. Once the button is pressed, the light will turn off.

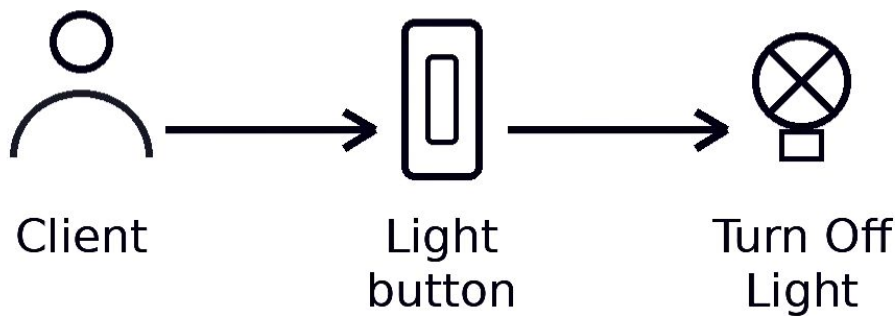


Figure 3.4 Switching the Light Off

If the client wants to turn on the ceiling fan, the client will locate the fan button and press the button to turn on the ceiling fan manually. Once the button is pressed, the ceiling fan will turn on.

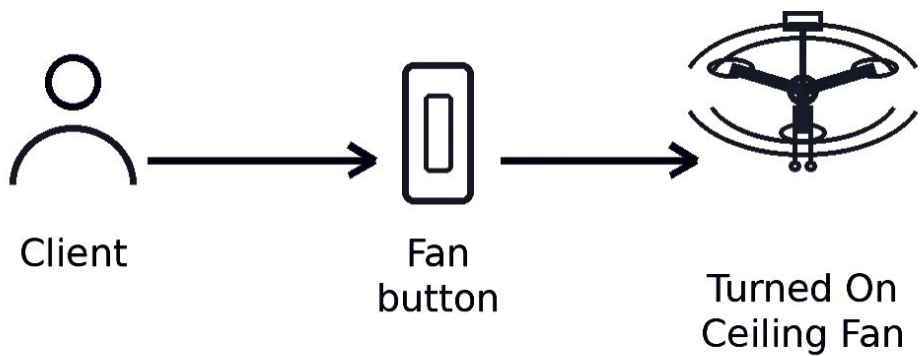


Figure 3.5 Switching the Ceiling Fan On

If the client wants to turn off the ceiling fan, the client will locate the fan button and press the button to turn off the ceiling fan manually. The string should be pulled first until the ceiling fan is not moving. After that, the button needs to be pressed and the ceiling fan will turn off.

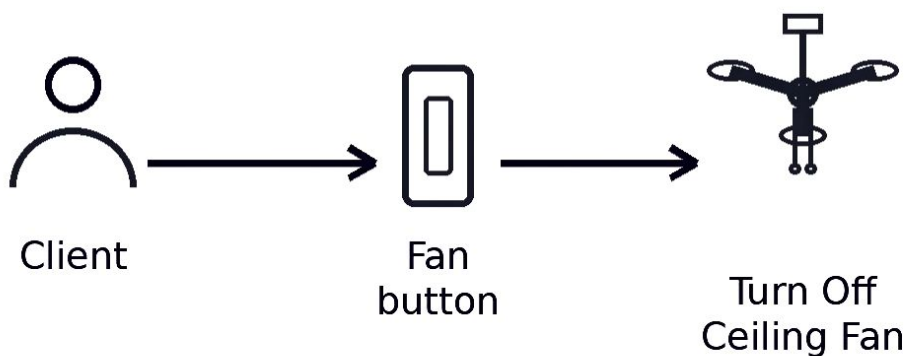


Figure 3.6 Switching the Ceiling Fan Off

If the client forgets to turn off the household appliances, the client needs to go back to their own house. The client needs to locate the different buttons of each household appliance so that the client can turn off each household appliance.

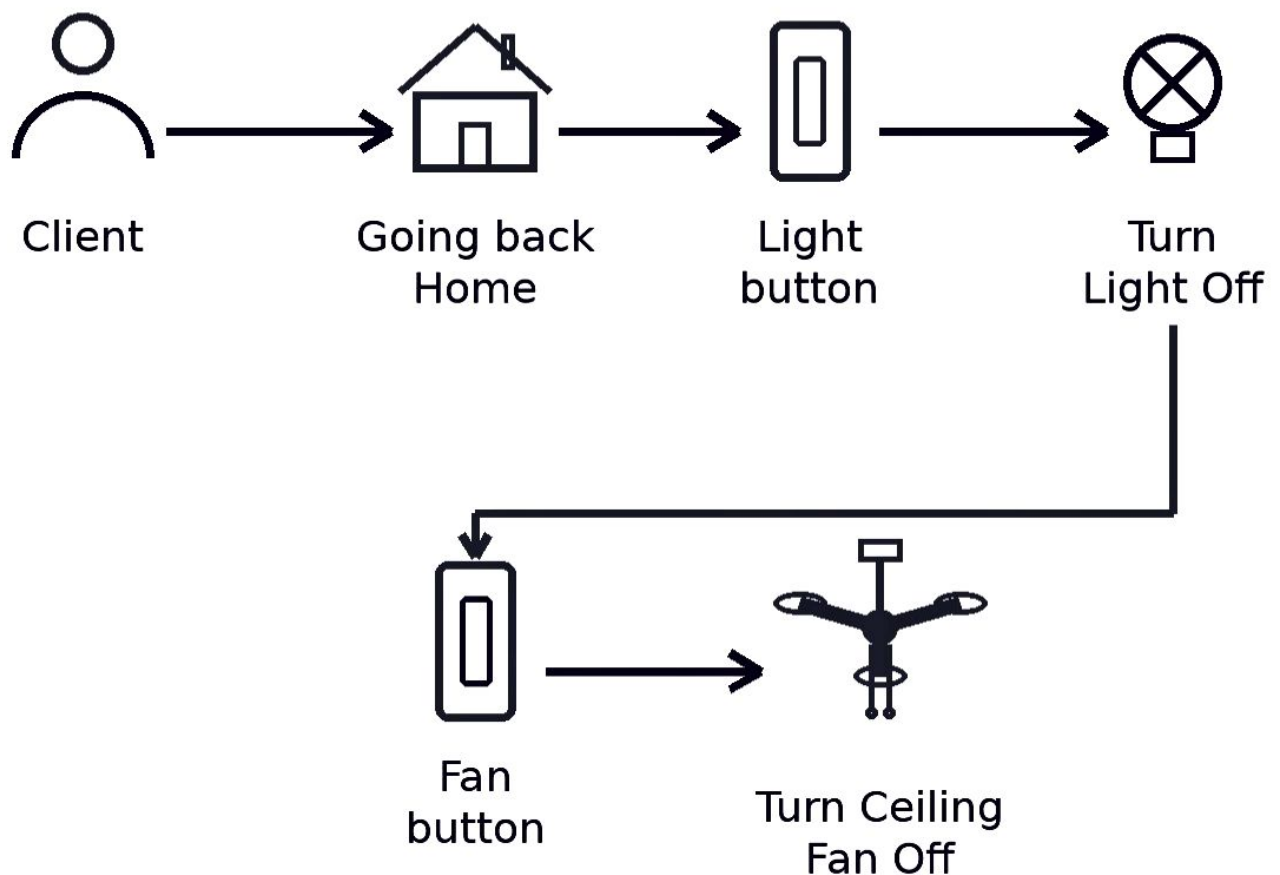
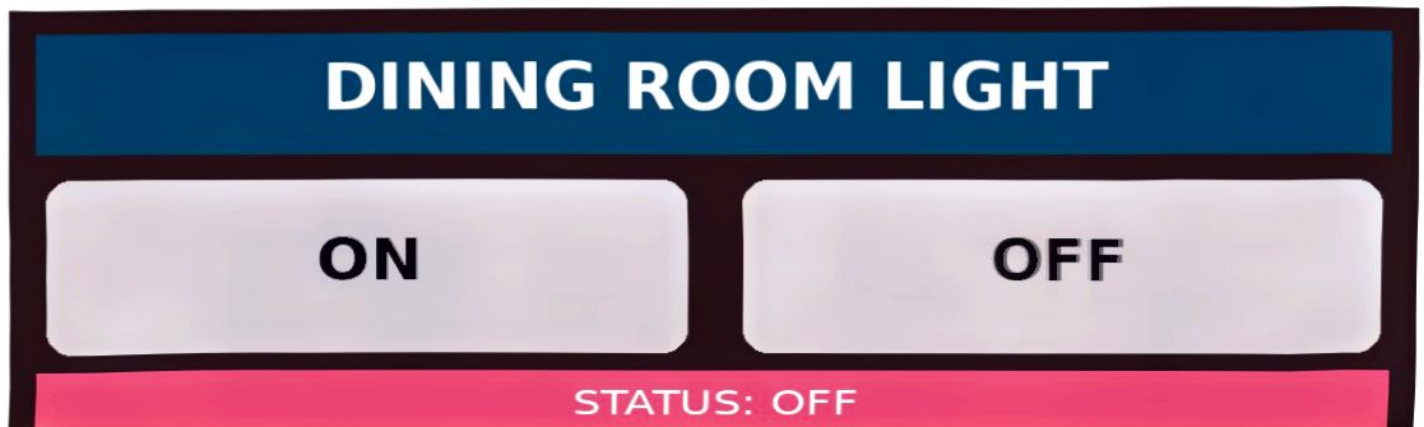


Figure 3.7 Switching the Household Appliances Off

The developers created a prototype and a system called the Home Automation Android app using Arduino.



If the smartphone is not connected to the Bluetooth module, the status of the kitchen light is off.



If the smartphone is not connected to the Bluetooth module, the status of the dining room light is off.



If the smartphone is not connected to the Bluetooth module, the status of the living room light is off.



If the smartphone is not connected to the Bluetooth module, the status of the bedroom 1 light is off.

BEDROOM 2 LIGHT

ON

OFF

STATUS: OFF

If the smartphone is not connected to the Bluetooth module, the status of the bedroom 2 light is off.

HALLWAY LIGHT

ON

OFF

GARAGE LIGHT

ON

OFF

STATUS: OFF

If the smartphone is not connected to the Bluetooth module, the status of the hallway light is off.

FANS

ALL FAN ON

ALL FAN OFF

FAN 1 ON

FAN 1 OFF

STATUS: OFF

FAN 2 ON

FAN 2 OFF

STATUS: OFF

FAN 3 ON

FAN 3 OFF

STATUS: OFF

the smartphone is not connected to the bluetooth module, the status of the fans are off.

GARAGE GATE

OPEN

CLOSED

STATUS: CLOSED

the smartphone is not connected to the bluetooth module, the status of the garage gate is closed.

HOME AUTOMATION

SELECT BLUETOOTH

DISCONNECT

NOT CONNECTED

The developers will be giving instructions to the homeowner on using the home automation application. The homeowner will press the select bluetooth button to gain access to the system. The homeowner will press the disconnect button when leaving the house.

Home Automation App

A0:AF:BD:C6:47:55 DESKTOP-5040MOT

00:1A:7D:DA:71:12 pc3

00:22:12:02:4F:0E HC-05

After pressing the select bluetooth button, the homeowner will connect to the bluetooth module with the corresponding bluetooth address and name.
Once the smartphone is connected to the bluetooth module, the homeowner can access the kitchen light and change the status to on.

KITCHEN LIGHT

ON

OFF

STATUS: ON

Once the smartphone is connected to the bluetooth module, the homeowner can access the dining light and change the status to on.

LIVING ROOM LIGHT

ON

OFF

STATUS: ON

Once the smartphone is connected to the bluetooth module, the homeowner can access the living room light and change the status to on.

BEDROOM 1 LIGHT

ON

OFF

STATUS: ON

Once the smartphone is connected to the bluetooth module, the homeowner can access the bedroom 1 light and change the status to on.

BEDROOM 2 LIGHT

ON

OFF

STATUS: ON

Once the smartphone is connected to the bluetooth module, the homeowner can access the bedroom 2 light and change the status to on.

HALLWAY LIGHT

ON

OFF

STATUS: ON

Once the smartphone is connected to the bluetooth module, the homeowner can access the hallway light and change the status to on.

GARAGE LIGHT

ON

OFF

STATUS: ON

Once the smartphone is connected to the bluetooth module, the homeowner can access the kitchen light and change the status to on.

FANS

ALL FAN ON

ALL FAN OFF

FAN 1 ON

FAN 1 OFF

STATUS: ON

FAN 2 ON

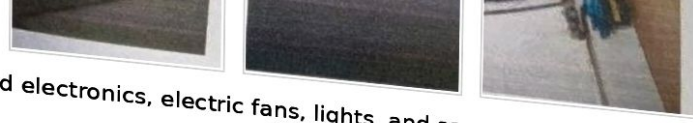
FAN 2 OFF

STATUS: ON

FAN 3 ON

FAN 3 OFF

STATUS: ON



Through the use of electricity and electronics, electric fans, lights, and servo motors convert electrical energy into mechanical or visual

Once the smartphone is connected to the bluetooth module, the homeowner can access the fans and change the status to on.

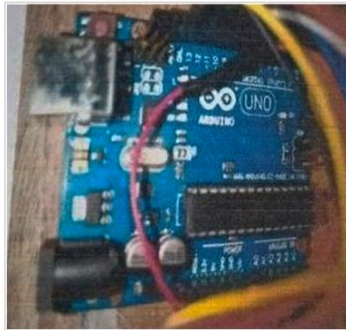
GARAGE GATE

OPEN

CLOSED

STATUS: OPEN

Once the smartphone is connected to the bluetooth module, the homeowner can access the garage gate and change the status to open.



The Arduino Uno microcontroller board is a useful tool for learning electronics and programming. It enables users to easily write

motion, enabling precise positioning, controlled airflow, and lighting in a variety of applications.

Implementation Plan

STRATEGY	ACTIVITIES	COMPONENTS NEEDED	PERSONS INVOLVED	DURATION
Define the scope and requirements of the app	Determine which devices needed for the system	Bluetooth module, smartphone	Homeowners	1-2 hours
Gather Materials	Acquire essential components and tools.	Arduino board, plywood, and nails	Homeowners	1-2 days
Design the Circuit	Create a circuit diagram for the system	Arduino, relay, power bank, jumper wires, servo motor, breadboard, Bluetooth module, fans, and lights	Developers	1-3 days
Write Arduino Code	Develop the software to control the system	Arduino IDE	Developers	2-6 days
Assemble Hardware	Build the physical components of the system.	Connect Bluetooth module, fans, lights, and servo motor	Homeowners and Developers	1-3 days
Testing and Validation	Verify the system performance	Conduct tests	Homeowners and Developers	1-2 weeks

CHAPTER 4

Summary, Conclusion, and Recommendation

Summary

The purpose of the study conducted by the developers, which resulted in the Home Automation Android App using Arduino, is to help ease the identified difficulties they encountered with the existing system. The development of a Home Automation Android App using Arduino was explored as a means to improve the convenience and efficiency of controlling various household appliances such as lights, fans, and garage door. The study aimed to provide a user-friendly and cost-effective solution, with a particular focus on assisting the elderly and disabled individuals.

This study the developers used Rapid Application Development which prioritized iterative development and prototyping to ensure the final product aligned with user requirements. The project's objectives included proper energy management, the development of a wireless home automation system controlled by an Android device, cost-effective system execution, and user-friendly design. Many tools for data analysis were also used in preparing the system and relating the relations of the data involved.

The following are the findings of the study:

1. The developers found out that the Home Automation Android App using Arduino is an effective and efficient means of controlling household appliances, providing convenience to users. This may have been supported by user feedback and data collected during the project.
2. This study may have found that the automation system contributes to cost saving by optimizing energy management, potentially reducing monthly utility bills for users. This would be a significant finding, especially for homeowners looking to lower their energy expenses.
3. This study may have identified that the Home Automation Android App using Arduino is particularly beneficial for the elderly and disabled individuals. It may have helped in simplifying daily tasks and providing more control over their living environment.
4. This study may have recognized that there is significant potential for further development and enhancement of home automation systems, indicating future research opportunities in this field.

Conclusion

With the details laid out and the findings about the status of the home automation system, the following are the conclusion of this study:

1. The development of the Home Automation Android App using Arduino provides a valuable solution for enhancing the convenience and comfort of controlling household appliances.
2. The study successfully met its objectives, including proper energy management, the creation of a Bluetooth typed automation system, cost-effectiveness, and user-friendliness.
3. The developers were able to create a responsive and adaptable system that aligns with user requirements.
4. The various groups, including homeowners and elderly or disabled individuals can benefit from the convenience and control offered by home automation.
5. Developers and future developers can build upon this study's foundation to further enhance home automation system and related studies.

Recommendation

With the findings and conclusions listed down, the following are recommended by the developers:

1. The developers should continue refining and enhancing the Home Automation Android App using Arduino to address potential and technical limitations and provide updates and new features to meet user needs.
2. The developers should consider providing user training and support resources to help individuals with varying levels of technical expertise to make the system more accessible to a wider audience.
3. Training of the expected users of the system will be held by the developers. This is to ensure the users will be familiar with the functions of the automation system.
4. Expand the range of compatible devices and technologies to make the system more versatile and adaptable to different household setups.
5. Collect and incorporate user feedback for ongoing system improvements. Regularly seek input from users to identify areas for enhancement and to address any issues and concerns.
6. Collaborate with organizations and institutions that focus on assisting elderly and disabled individuals to provide tailored solutions that cater to their specific needs.
7. Encourage the future developers to explore different technologies to improve the automation system's capabilities.

Kumar et. al. (2022, May). Bluetooth Based Home Automation. *International Journal of Engineering Research in Electronics and Communication Engineering (IJERECE)*.
https://www.technoarete.org/common_abstract/pdf/IJERECE/v9/i5/Est_93615.pdf

M

Malik et. al. (2020, November). A Home Automation Through Android Mobile App by Using Arduino UNO. *ResearchGate*.
https://www.researchgate.net/publication/345992939_A_Home_Automation_Through_Android_Mobile_App_by_Using_Arduino_UNO

Malik et. al. (2022, April). Design of Smart Home System Using Arduino. *Mathematical Statistician and Engineering Applications*.
<https://www.google.com/url?sa=t&source=web&rct=j&opi=899784&url=https://www.philstat.org/index.php/MSEA/article/download/1466/973/2490&ved=2ahUKEwizg6yBvKKDAXrd2wQHbWcBloQFnocBgQAQ&usq=AOyVaw1CcLa-ZsnjezqswVDKnobH>

Morallo, Narcisa (2017). Designing of Lighting Automation System Based on Arduino Bluetooth Interface using Android Smartphone as Platform. *ejournal.ph*.
<https://www.google.com/url?sa=t&source=web&rct=j&opi=899784&url=https://ejournals.ph/article.php%3Fid%3D12301&ved=2ahUK>

[w7d3LakwFu0r-di5lc](#)

N

Ngerem et. al. (2020, November). A Home Automation System Based on *Bluetooth Technology Using an Android Smartphone*. *ResearchGate*.

https://www.researchgate.net/publication/346450562_A_Home_Automation_System_Based_on_Bluetooth_Technology_Using_an_Android_Smartphone

O

Oo et. al. (2018). *Bluetooth Based Home Automation System Using Android and Arduino*. *Academia*.

https://www.academia.edu/39096302/Bluetooth_Based_Home_Automation_System_Using_Android_and_Arduino

P

Piyare, Rajeev (2011, June). *Bluetooth Based Home Automation using Cellphone*. *ResearchGate*.

https://www.researchgate.net/publication/231182479_Bluetooth_based_home_automation_system_using_cell_phone

R

Rosendahl et. al. (2007, July). *A Home Automation Through Android Mobile App by Using Arduino Uno*. *ResearchGate*.

https://www.researchgate.net/publication/221303243_Mobile_Home_Automation_-

[_Merging_Mobile_Value_Added_Services_and_Home_Automation_Technologies](#)

S

Selvaraj, Kannapiran (2016, October). A Novel Home Automation System using Bluetooth and Arduino. *ResearchGate*.

https://www.researchgate.net/publication/313672508_A_Novel_Home_Automation_System_using_Bluetooth_and_Arduino

Singh et. al. (2021, June). Arduino Based Home Automation Using Bluetooth Android Smartphone. *International Journal of Research in Engineering and Technology*.

<https://www.impactjournals.us/index.php/download/archives/03-07>

APPENDIX A
Survey Questionnaire
GOLDEN WEST COLLEGE

College of Information Technology Education • Alaminos City,
Pangasinan

Name: Lorenz Jesus B. Date: _____

Address: Binmaley, Pangasinan

1. Do you have appliances in your home, if have what it is?

Yes - TV, refrigerator, electric fan, DVD player, rice cooker

2. How often do you use your appliances mentioned in above?

Everyday

3. How important that we switch off some appliances that we didn't use?

To save energy

4. Are you familiar with smart home devices?

No

5. Have you ever used any home automation devices?

No

6. If you already encounter home automation, what challenges have you faced while using home automation systems? If yes, please state it.

Not encountered yet

7. What are your main reasons for considering home automation?

For convenient and easy living

8. What are the factors influence your decision to adopt or switch to a new home automation system?

To make things easier and to discover new things

9. How would rate the reliability and responsiveness of the home automation system?

N/A

10. What features or functionalities would you like to see improved or added in your home automation?

To have voice control

APPENDIX B
Sample Survey
GOLDEN WEST COLLEGE

**College of Information Technology Education • Alaminos City,
Pangasinan**

BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY
RESEARCH PROJECT ACCEPTABILITY TEST SURVEY

Dear Sir / Ma'am,

We appreciate your participation in this survey to evaluate the acceptability and usability of our Bachelor of Science in Information Technology research project, titled "HOME AUTOMATION ANDROID APP USING ARDUINO".

Your feedback is invaluable in helping us understand the effectiveness and practicality of our project.

Please take a few minutes to complete this survey honestly and to the best of your knowledge. Your responses will remain confidential, and your identity will not be disclosed.

Part 1: Personal Information (Optional)

Name: _____ [Optional]

Email (if you'd like to receive project updates):
_____ [Optional]

Part 2: Project Introduction

Have you been introduced to "[HOME AUTOMATION ANDROID APP USING ARDUINO]"?

☒ Yes ☐ No

Please provide your initial impressions or expectations of this project:

Very useful

Part 3: Usability and Functionality

Have you had the opportunity to interact with the "[HOME AUTOMATION ANDROID APP USING ARDUINO]"?

☒ Yes ☐ No

How would you rate the usability and functionality of the "[HOME AUTOMATION ANDROID APP USING ARDUINO]"?

GOLDEN WEST COLLEGE

**College of Information Technology Education • Alaminos City,
Pangasinan**

- ☒ Excellent ☐ Good ☐ Fair
☐ Poor ☐ Very Poor

If you have used the system, please describe your experience and any challenges you encountered.

Easy to use

Part 4: Relevance and Practicality

In your opinion, how relevant is the “[HOME AUTOMATION ANDROID APP USING ARDUINO]” system to addressing real-world challenges?

- ☐ Highly Relevant ☒ Relevant
☐ Somewhat Relevant ☐ Not Relevant

Do you believe that the implementation of the “[HOME AUTOMATION ANDROID APP USING ARDUINO]” system is practical and feasible?

- ☐ Strongly Agree ☒ Agree
☐ Neutral ☐ Disagree
☐ Strongly Disagree

Part 5: Impact and Improvement

What potential impact do you see in the widespread adoption of the “[HOME AUTOMATION ANDROID APP USING ARDUINO]”?

Easy access and can save energy consumption

Are there any specific suggestions or improvements you would recommend for the “[HOME AUTOMATION ANDROID APP USING ARDUINO]”?

Nothing

Part 6: Overall Acceptability

Overall, how acceptable do you find the “[HOME AUTOMATION ANDROID APP USING ARDUINO]” research project?

- ☐ Highly Acceptable ☒ Acceptable
☐ Neutral ☐ Unacceptable
☐ Highly Unacceptable

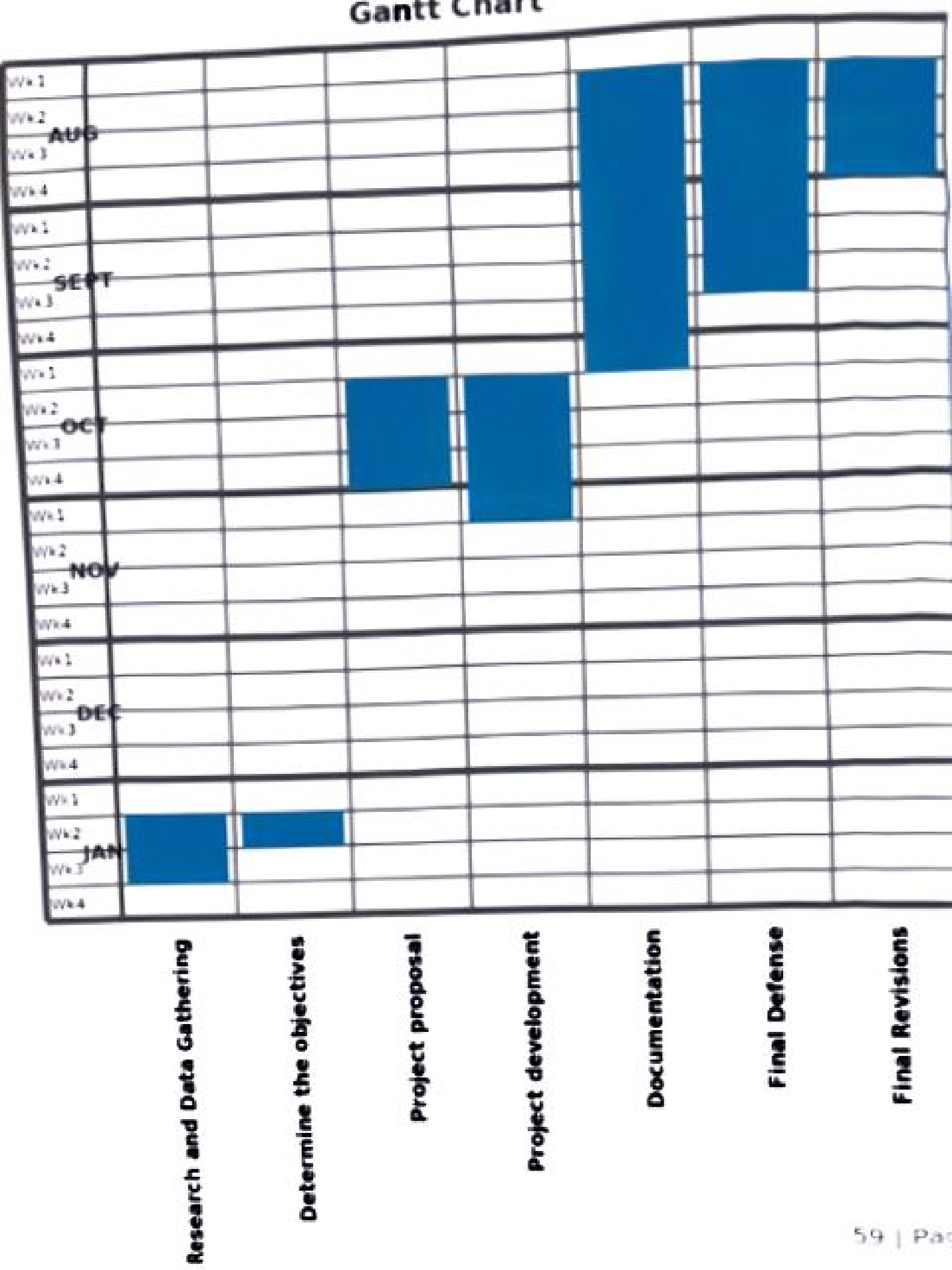
GOLDEN WEST COLLEGE

**College of Information Technology Education • Alaminos City,
Pangasinan**

Thank you for your participation in this survey. Your feedback is essential for us to evaluate and enhance the “[HOME AUTOMATION ANDROID APP USING ARDUINO]” research project. If you have any additional comments or concerns, please feel free to contact us.

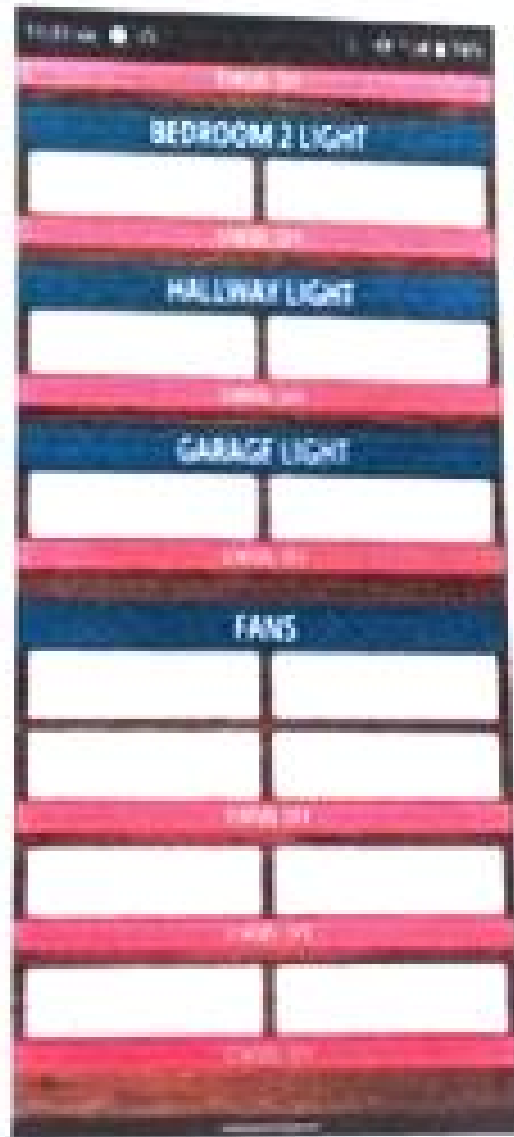
SIGNATURE

Gantt Chart



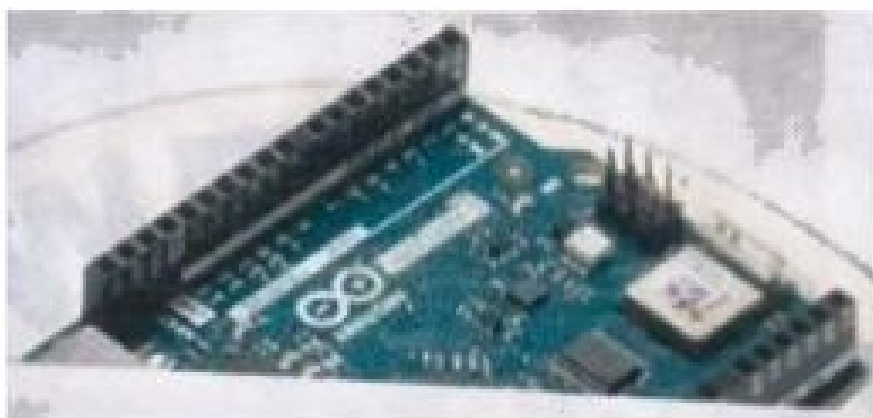


<https://drive.google.com/drive/folders/1Bwsc52Cv6w...>
[snPNPZSSn?usp=sharing](https://drive.google.com/drive/folders/1Bwsc52Cv6w...)



Arduino Uno. This will be the brain of the system. It is responsible for controlling and coordinating various components. It can read inputs and control outputs for activating the system.

Plate 2. Arduino Uno



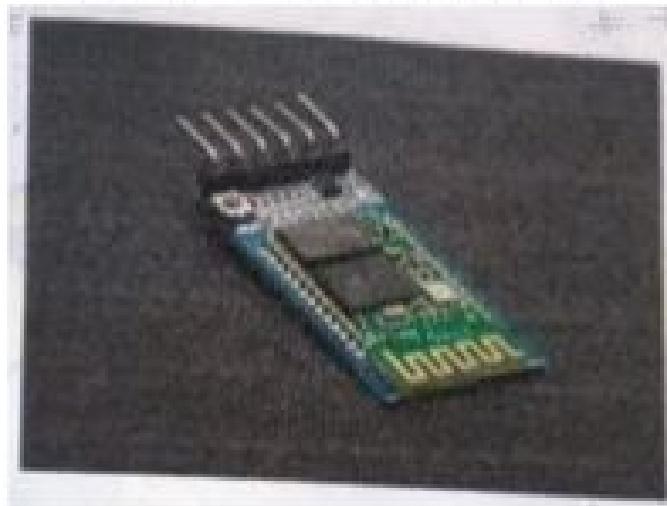
Servo Motor. It is used for controlling the movement of the garage door which it has a precise control of angular position.

Plate 3. Servo Motor



HC - 05 Bluetooth Module. It enables wireless communication between the Arduino and the Android device of the client. This allows to control the home automation system using the MIT Application.

Plate 4. HC - 05 Bluetooth Module



Relay. It acts as an electrically operated switch, allowing the Arduino to control high-voltage devices like lights or appliances. It is also the crucial component for interfacing the electrical systems safely.

Plate 5. Relay



Jumper Wires. create electrical connections between different components on the breadboard or between components and the Arduino. It is also providing a convenient way to prototype circuits.



Breadboard. platform for quickly prototyping and testing circuits with soldering. It allows to connect components using jumper wires.

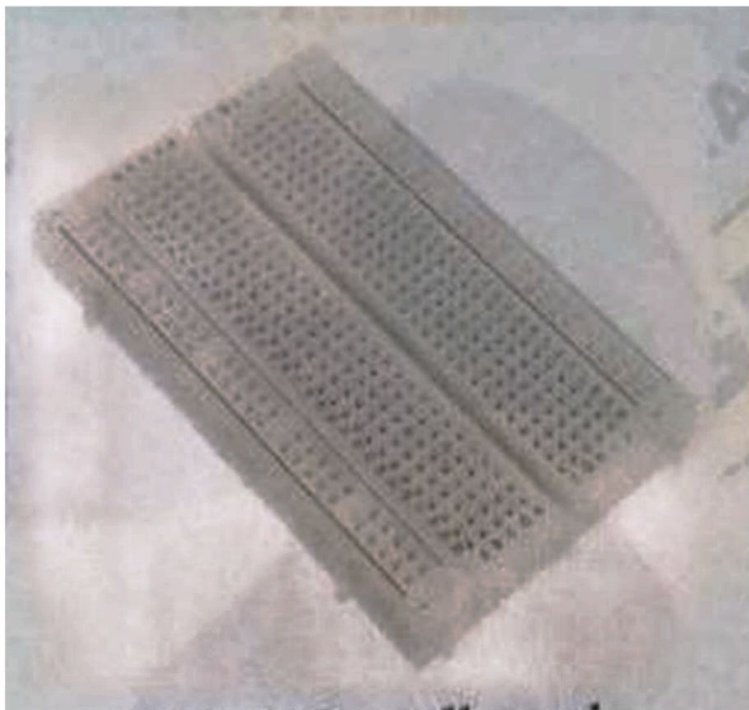


Plate 7. Breadboard

Module 8. Direct Current Motor

*fans. n be controlled by the Arduino to regulate temperature
air circulation of a room.*

LED Lights. as visual indicators or controllable light source in the home automation system.



Plate 10. LED Lights

Power Bank / Power Source. rduino and other components which it makes the system portable and not reliant on a constant power source.

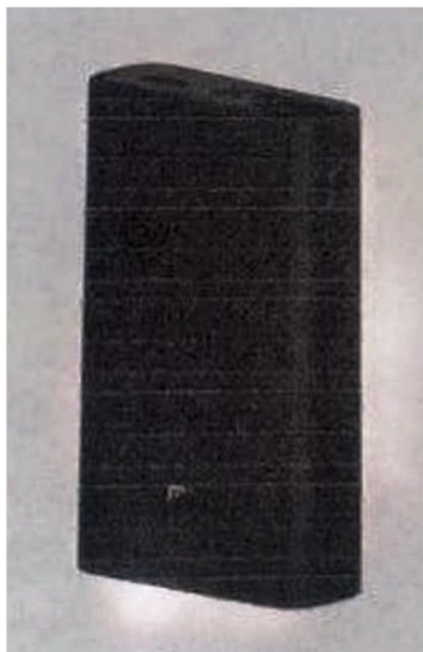


Plate 11. Power Bank / Power Source

al Serial Bus. g the Arduino to a
r/laptop for programming and debugging.



Plate 12. Universal Serial Bus

CALONGE, JEHOSHAPHAT DELA PEÑA

Purok 3 Brgy. Calzada, Mabini, Pangasinan

09386852523

jehoshaphatcalonge@gmail.com



EDUCATIONAL BACKGROUND

COLLEGE

- Bachelor of Science in Information Technology
- Golden West Colleges
- Poblacion, Alaminos City, Pangasinan
- 2020-2024

SENIOR HIGH SCHOOL

- St. Adelaide School of the Philippines
- Burgos, Pangasinan
- 2020-2021

JUNIOR HIGH SCHOOL

- Tagudin National High School
- Tagudin, Mabini, Pangasinan
- 2015-2019

ELEMENTARY

- Mabini Central School
- Poblacion, Mabini, Pangasinan
- 2009-2015

PIOQUINTO, JEREMY LEGASPI

Lucap, Alaminos City, Pangasinan

09076523155

jeremypioquinto1@gmail.com



EDUCATIONAL BACKGROUND

COLLEGE

- Bachelor of Science in Information Technology
- Golden West Colleges
- San Jose Drive, Poblacion, Alaminos City, Pangasinan
- 2020-2024

SENIOR HIGH SCHOOL

- Golden West Colleges
- San Jose Drive, Poblacion, Alaminos City, Pangasinan
- 2019-2021

JUNIOR HIGH SCHOOL

- Inerangan National High School
- Inerangan, Alaminos City, Pangasinan
- 2015-2019

ELEMENTARY

- Lucap Elementary School
- Lucap, Alaminos City, Pangasinan
- 2009-2015

CARDENAS, ANGEL ROYO

San Vicente, Alaminos City, Pangasinan

0992 304 5194

ar20cardenasu@gmail.com

EDUCATIONAL BACKGROUND

COLLEGE

- Bachelor of Science in Information Technology
- Golden West Colleges
- San Jose Drive, Alaminos City, Pangasinan
- 2020-2024

SENIOR HIGH SCHOOL

- Alaminos City National High School
- Magsaysay, Alaminos City, Pangasinan
- 2019-2021

JUNIOR HIGH SCHOOL

- Alaminos City National High School
- Magsaysay, Alaminos City, Pangasinan
- 2015-2019

ELEMENTARY

- San Vicente Elementary School
- San Vicente, Alaminos City, Pangasinan
- 2009-2015

REYES, MATTHEW BESENIO

*001 Zone IV San Vicente, Alaminos City,
Pangasinan*

09612272640

matthewbesenioreyes@gmail.com



EDUCATIONAL BACKGROUND

COLLEGE

- Bachelor of Science in Information Technology
- Golden West Colleges
- Poblacion, Alaminos City, Pangasinan
- 2020-2024

SENIOR HIGH SCHOOL

- The Great Plebeian College
- Poblacion, Alaminos City, Pangasinan
- 2016 - 2018

JUNIOR HIGH SCHOOL

- The Great Plebeian College
- Poblacion, Alaminos City, Pangasinan
- 2012 - 2016

ELEMENTARY

- San Vicente Elementary School
- San Vicente, Alaminos City, Pangasinan
- 2006 - 2012

TAGARA, PHINKIE RICO

Brgy. Garrita Bani Pangasinan

09305570709

p.tagara3@gmail.com



EDUCATIONAL BACKGROUND

COLLEGE

- Bachelor of Science in Information Technology
- Golden West Colleges
- San Jose Drive, Alaminos City, Pangasinan
- 2020-2024

SENIOR HIGH SCHOOL

- The Great Plebeian College
- Don Reinoso St., Poblacion, Alaminos City, Pangasinan
- 2019-2021

JUNIOR HIGH SCHOOL

- Garrita National High School
- Garrita, Bani, Pangasinan
- 2015-2019

ELEMENTARY

- Abunciang Elementary School
- Sitio Abunciang, San Miguel, Bani, Pangasinan
- 2009-2015

DAYRIT, MELANIE VILAR

Poblacion East Agno Pangasinan

09637378212

mheldayrit2@gmail.com



EDUCATIONAL BACKGROUND

COLLEGE

- Bachelor of Science in Information Technology
- Golden West Colleges
- Poblacion, Alaminos City, Pangasinan
- 2020-2024

SENIOR HIGH SCHOOL

- Agno National High School
- Poblacion West, Agno, Pangasinan
- 2019-2021

JUNIOR HIGH SCHOOL

- Agno National High School
- Poblacion West, Agno, Pangasinan
- 2015-2019

ELEMENTARY

- Agno Central School
- Poblacion West, Agno, Pangasinan
- 2009-2015

BRAVO, JERICHO PANTUA

Sitio Cuala, Tagudin, Mabini, Pangasinan

09673005614

bravojericho74@gmail.com



EDUCATIONAL BACKGROUND

COLLEGE

- Bachelor of Science Information Technology
- Golden West Colleges
- Poblacion, Alaminos City, Pangasinan
- 2020-2024

SENIOR HIGH SCHOOL

- Tagudin National High School
- Tagudin, Mabini, Pangasinan
- 2019-2021

JUNIOR HIGH SCHOOL

- Tagudin National High School
- Tagudin, Mabini, Pangasinan
- 2015-2019

ELEMENTARY

- Mabini Central School
- Poblacion, Mabini, Pangasinan
- 2009-2015